

Folio Software Requirements Specification

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1 Introduction

The purpose of this document is to define the functional, interfaces, and functional behavior requirements for Folio as a system. These requirements will guide Folio's iterative development as a reference for software testing and implementation. Folio is an academic planning tool that allows students to create a comprehensive study schedule and encourages them to complete tasks. The major capabilities of the system are academic planning, coursework organization, task scheduling, instructor feedback reception, CSV/manual course data entry, progress tracking, optional gamification features, and plan exports. The primary users are students who can use Folio for independent study planning, with optional game/reward features to encourage them to curate schedules and complete tasks.

2 Functional Requirements

2.1 Coursework & Plan Creation

F1. The system shall allow the student to select an assignment, test, or other coursework to address.

F2. The system shall allow the student to choose a planning template.

F3. The system shall allow the student to edit suggested tasks.

F18. The system shall allow the student to manually create a coursework record that includes, at a minimum, a course, an assignment name, and a due date.

F19. The system shall allow the student to edit a manually entered coursework record after it has been created.

F20. The system shall allow the student to delete a coursework record.

F21. The system shall validate manually entered coursework before saving the coursework record.

F22. The system shall reject a manually entered coursework record when the course or assignment name is blank, the due date is missing, or the due date does not match the date format specified by the manual-entry interface.

F23. The system shall identify each field that prevents a manually entered coursework record from being saved.

2.2. Instructor Feedback & Plan Revision

F4. The system shall allow the student to accept an instructor's suggestion.

F5. The system shall allow the student to reject an instructor's suggestion.

F6. The system shall allow the student to adapt an instructor's suggestion before applying it to the plan.

F7. The system shall allow the student to decide whether to revise the plan after instructor feedback.

2.3 CSV Import & Validation

F24. The system shall allow the student to import coursework from a CSV file that contains the required Folio coursework fields.

F25. The system shall validate each CSV record before saving the imported coursework record.

F26. The system shall reject a CSV record that omits a required field or contains a value that does not conform to the data type or format specified for that field by the CSV import interface.

F27. For each rejected CSV record, the system shall report the record's location in the CSV file and each reason for rejection.

F28. The system shall reject a CSV file that cannot be parsed or does not contain the required columns without adding any records from that file.

F29. The system shall treat two CSV records as duplicates when all required coursework fields have identical values and shall add no more than one of those records during a single CSV import.

F30. The system shall import every valid CSV record when other records in the same CSV file are invalid.

2.4 Data Persistence and Relationships

F31. The system shall persist coursework records, academic plans, scheduled tasks, and game and reward states across application sessions.

F32. The system shall restore the student's most recently saved coursework records, academic plans, scheduled tasks, and game and reward state when the student begins a subsequent application session.

F33. The system shall preserve each association between a scheduled task and the coursework record or academic plan to which it belongs.

2.5 Scheduling & Task Management

- F8. The system shall provide a daily view of scheduled tasks.
- F9. The system shall provide a weekly view of scheduled tasks.
- F10. The system shall allow the student to reschedule a task.
- F11. The system shall allow the student to configure reminders.
- F12. The system shall allow the student to mark a task complete.
- F34. The system shall display scheduled tasks from multiple academic plans in a combined schedule.
- F35. The system shall preserve scheduled tasks from separate academic plans as distinct records in the combined schedule.
- F40. The system shall update the combined schedule when the date or time of a scheduled task changes.
- F41. The system shall remove a scheduled task from the combined schedule when the student deletes it from their academic plan.

2.6 Scheduling Conflict Detection & Resolution

- F36. The system shall detect a scheduling conflict when one scheduled task starts before another scheduled task ends and ends after the other scheduled task starts.
- F37. The system shall notify the student when creating or rescheduling a task that conflicts with an existing scheduled task.
- F38. The system shall identify every scheduled task involved in a detected scheduling conflict.
- F39. The system shall allow the student to resolve a scheduling conflict by rescheduling at least one of the conflicting tasks.

2.7 Progress, Game & Reward Behavior

- F13. The system shall show a progress bar for each plan.
- F14. The system shall provide a congratulatory response when the student completes a task.
- F42. The system shall count a scheduled task toward academic plan progress only while it is marked complete.
- F43. When a scheduled task with an associated reward changes from incomplete to complete, the system shall award the associated reward no more than once for that status transition.
- F44. The system shall update the progress of an academic plan when the completion status of a scheduled task in that plan changes.
- F45. The system shall calculate an academic plan's progress using only the scheduled tasks that belong to that academic plan.
- F46. The system shall not modify coursework records when game or reward data changes.

F47. The system shall exclude game and reward data from academic information prepared for sharing with an instructor.

F48. Disabling the game and reward features shall not delete or alter the student's academic plans, coursework records, or scheduled tasks.

2.8 Independent & Optional Workflows

F15. The system shall allow the student to use the planning workflow without instructor participation.

F16. The system shall allow the student to use the planning workflow without the game or rewards.

F17. The system shall use course information as planning context without inferring studying, attendance, participation, or effort outside the system.

F49. The system shall allow the student to create and save an academic plan without sharing information with an instructor.

F50. The system shall allow the student to modify and complete an academic plan without instructor feedback.

F51. The system shall allow the student to schedule, reschedule, and complete scheduled tasks without instructor participation.

F52. The absence of instructor feedback shall not prevent the student from viewing an existing academic plan.

3 Interface Requirements

3.1 Academic Export & Instructor-Sharing Interfaces

I1. The system shall export a plan as a PDF.

I2. The system shall export a plan as a DOCX.

I3. The system shall prepare an instructor's email and present it to the student for review before sending.

I4. The system shall allow the student to choose which academic plan information is prepared for instructor sharing.

I5. The system shall exclude progress, reward, and game information from academic exports and prepared instructor emails.

3.2 CSV Import & Manual Entry Interfaces

- I6. The system shall accept coursework imports in CSV format.
- I7. The CSV import interface shall identify all required columns and the accepted data format for each column before the student initiates an import.
- I8. After processing a CSV import, the system shall display the number of accepted and rejected CSV records.
- I9. For each rejected CSV record, the CSV import interface shall display the record's location in the file and each validation error.
- I12. The manual-entry interface shall provide a labeled input control for each field required to create a valid coursework record.
- I13. The manual-entry interface shall specify the accepted date format for a coursework record's due date.
- I14. When the system rejects a manually entered coursework record, the manual-entry interface shall identify each field that caused the rejection and explain the applicable validation rule.

3.3 Scheduling Interface

- I10. The scheduling interface shall display the name of the associated course or academic plan for each scheduled task in the combined schedule.
- I11. The scheduling interface shall display a conflict indicator on each scheduled task involved in a detected scheduling conflict.

4 Performance Requirements

- P1. The system shall preserve the student's entered and edited plan information when the plan is saved or revised.
- P2. PDF, DOCX, and prepared email outputs shall accurately contain the academic plan information selected for sharing.
- P3. The displayed task-completion and plan-progress information shall match the student's saved task-completion status.
- P4. The system shall display the requested stored academic plan within 2 seconds of the student submitting the request.
- P5. The system shall update the displayed schedule within 2 seconds of the student creating, modifying, or deleting a scheduled task.
- P6. The system shall detect and display all scheduling conflicts within 2 seconds of the student adding or rescheduling a task.
- P7. The system shall import and validate a CSV file containing 500 coursework records within 5 seconds after the student initiates the import.

P8. The system shall confirm that it has saved a change to a coursework record, academic plan, or scheduled task within 2 seconds of the student submitting the change.